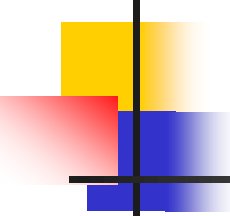


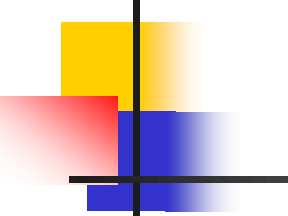
第二章作业 答案



T1

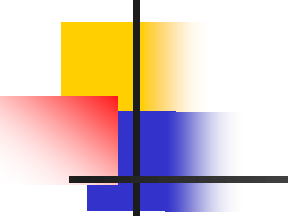
Simplify the function into the simplest “AND-OR” form.

$$F = \overline{\overline{A}BC} + \overline{A}BCD + \overline{A}E$$


$$\begin{aligned} F &= \overline{\overline{ABC}} + \overline{ABCD} + \overline{AE} \\ &= \overline{\overline{ABC}} + D + \overline{AE} \\ &= \overline{A} + B + \overline{C} + D + \overline{AE} \\ &= \overline{A} + B + \overline{C} + D \end{aligned}$$

Simplify the function into the simplest “OR-AND” form.

$$F = (A + B + E) \cdot (A + \overline{B} + D) \cdot (A + D + E)$$

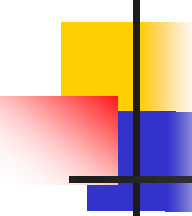


$$\begin{aligned}
 F^D &= ABE + \overline{A}BD + ADE \\
 &= A(BE + \overline{B}D + DE) \\
 &= A(BE + \overline{B}D) \\
 &= ABE + \overline{A}BD
 \end{aligned}$$

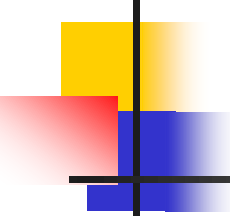
$$(F^D)^D = (A + B + E) \bullet (A + \overline{B} + D)$$

Simplify the following function into simplest AND-OR form by algebraic method.

$$\overline{A} \overline{C} + \overline{A} D + \overline{A} C \overline{D} + A \overline{C} \overline{D} + ABD + A \overline{B}$$



$$\begin{aligned}
 & \overline{A} \cdot \overline{C} + \overline{A} \cdot D + \overline{A} \cdot C \cdot \overline{D} + A \cdot \overline{C} \cdot \overline{D} + A \cdot B \cdot D + A \cdot \overline{B} \\
 &= \overline{A}(\overline{C} + D + C \cdot \overline{D}) + A(\overline{C} \cdot \overline{D} + B \cdot D + \overline{B}) \\
 &= \overline{A}(\overline{C} + D + \overline{D}) + A(\overline{C} \cdot \overline{D} + D + \overline{B}) \\
 &= \overline{A}(\overline{C} + 1) + A(\overline{C} + D + \overline{B}) \\
 &= \overline{A} + A(\overline{C} + D + \overline{B}) \\
 &= \overline{A} + \overline{B} + \overline{C} + D
 \end{aligned}$$

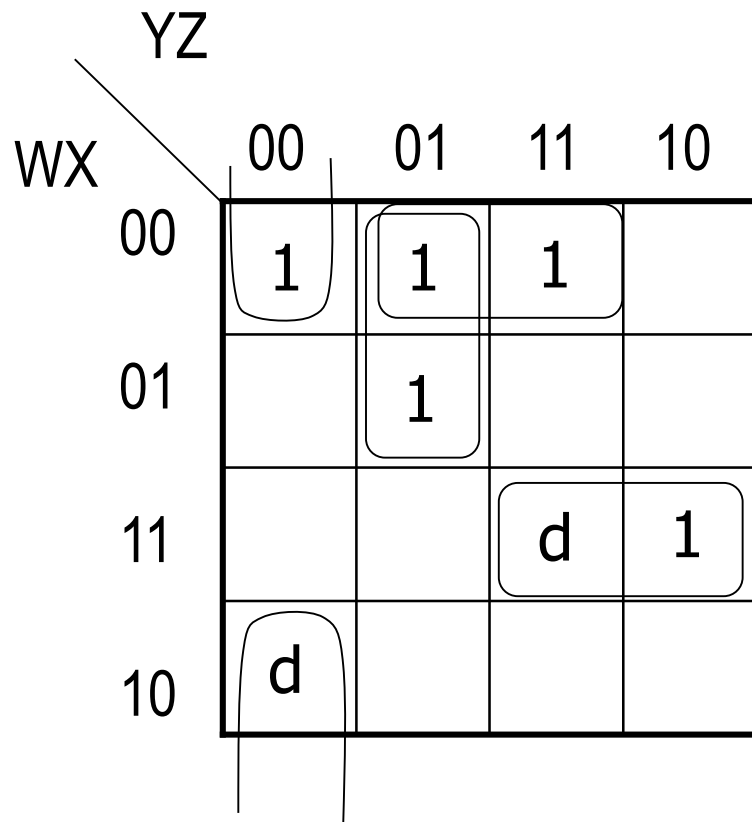


T4 - Two Solutions

Simplify the function with K-map, and write the simplest “AND-OR” function.

$$(a) F = \sum_{w,x,y,z} (0,1,3,5,14) + d(8,15)$$

Simplify the function with K-map, and write the simplest “AND-OR” function.



$$F = \bar{X}\bar{Y}\bar{Z} + \bar{W}\bar{Y}Z + \bar{W}\bar{X}Z + WXY$$

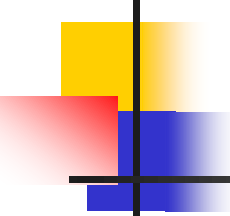
Solution One

Simplify the function with K-map, and write the simplest “AND-OR” function.

YZ WX					
		00	01	11	10
00	1	1	1		
01		1			
11			d	1	
10	d				

$$F = \bar{W}\bar{X}\bar{Y} + \bar{W}\bar{Y}Z + \bar{W}\bar{X}Z + WXY$$

Solution Two

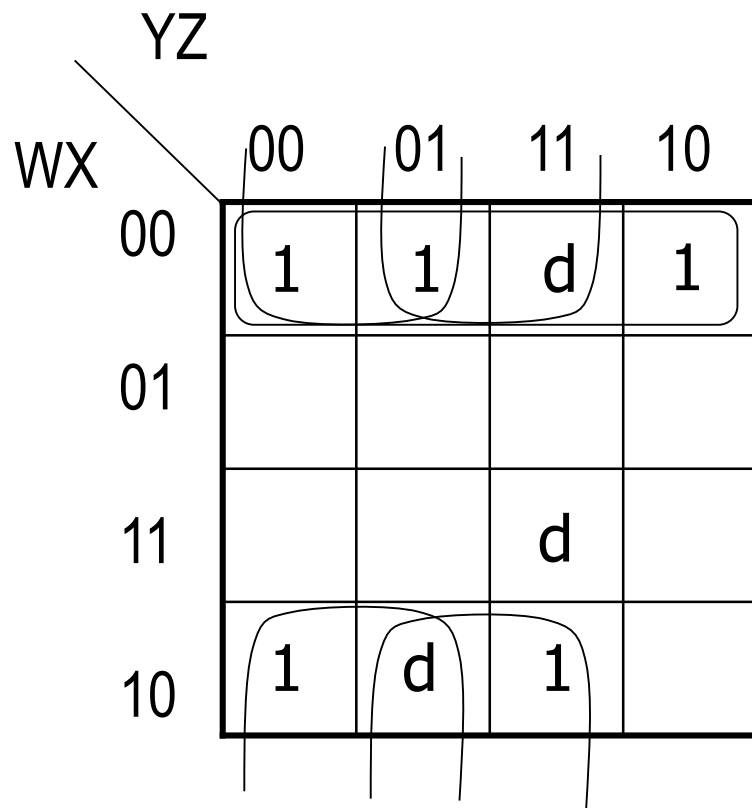


T5

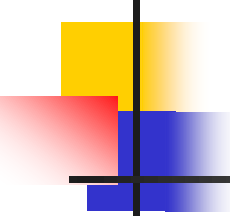
Simplify the function with K-map, and write the simplest “AND-OR” function.

$$(b) F = \sum_{w,x,y,z} (0,1,2,8,11) + d(3,9,15)$$

Simplify the function with K-map, and write the simplest “AND-OR” function.



$$F = \bar{X}\bar{Y} + \bar{X}Z + \bar{W}\bar{X}$$



T6-Two Solutions

Simplify the function with K-map, and write the simplest “AND-OR” function.

$$(c)F = \sum_{A,B,C,D} (4, 6, 7, 9, 13) + d(12)$$

Simplify the function with K-map, and write the simplest “AND-OR” function.

		CD			
		00	01	11	10
AB	00				
	01	1		1	1
	11	d	1		
	10		1		

$$F = B\bar{C}\bar{D} + A\bar{C}D + \bar{A}BC$$

Solution One

Simplify the function with K-map, and write the simplest “AND-OR” function.

		CD			
		00	01	11	10
AB	00				
	01	1		1	1
	11	d	1		
	10		1		

$$F = \bar{A}\bar{B}\bar{D} + A\bar{C}D + \bar{A}BC$$

Solution Two

Simplify the function with K-map, and write the simplest “AND-OR” function.

$$(d)F = \sum_{A,B,C,D} (1, 5, 12, 13, 14, 15) + d(7, 9)$$

Simplify the function with K-map, and write the simplest “AND-OR” function.

		CD			
		00	01	11	10
AB	00		1		
	01		1	d	
	11	1	1	1	1
	10		d		

$$F = AB + \bar{C}D$$